

Exercise-Introduction-to-Database

```
##Create Schema
CREATE DATABASE store;

## 1- create table countries
CREATE TABLE countries (
    code INT PRIMARY KEY,
    name VARCHAR(20),
    continent_name VARCHAR(20)
);

## 2- create table users
CREATE TABLE users (
    id INT PRIMARY KEY,
    full_name VARCHAR(20),
    email VARCHAR(20),
    gender CHAR(1),
    date_of_birth VARCHAR(15),
    created_at DATETIME,
    country_code INT,

    FOREIGN KEY (country_code)
        REFERENCES countries(code)
);

## 3- create table product
CREATE TABLE products (
    id INT PRIMARY KEY,
    name VARCHAR(10),
    price INT,
    status VARCHAR(10),
    created_at DATETIME
);

## 4- create table orders
CREATE TABLE orders (
    id INT PRIMARY KEY,
    user_id INT,
    status VARCHAR(6),
    created_at DATETIME,

    FOREIGN KEY (user_id)
        REFERENCES users(id)
);

## 5- order_products
CREATE TABLE order_products (
    order_id INT,
    product_id INT,
    quantity INT,

    PRIMARY KEY (order_id, product_id),

    FOREIGN KEY (order_id)
        REFERENCES orders(id),
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```
        FOREIGN KEY (product_id)
            REFERENCES products(id)
    );

## <-----Q3----->
-- countries

ALTER TABLE countries
ADD CONSTRAINT countries_name
UNIQUE (name);

ALTER TABLE countries
MODIFY continent_name VARCHAR(20) NOT NULL;

-- users

ALTER TABLE users
ADD CONSTRAINT uq_users_email
UNIQUE (email);

ALTER TABLE users
ADD CONSTRAINT chk_users_gender
CHECK (gender IN ('m', 'f'));

-- orders

ALTER TABLE orders
ADD CONSTRAINT chk_orders_status
CHECK (status IN ('start', 'finish'));

-- order_products

ALTER TABLE order_products
MODIFY quantity INT DEFAULT 0;

-- products

ALTER TABLE products
MODIFY price INT DEFAULT 0;

ALTER TABLE products
MODIFY name VARCHAR(10) NOT NULL;

ALTER TABLE products
ADD CONSTRAINT chk_products_status
CHECK (status IN ('valid', 'expired'));

-- created_at

ALTER TABLE users
MODIFY created_at DATETIME DEFAULT CURRENT_TIMESTAMP;
```

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```
ALTER TABLE orders
MODIFY created_at DATETIME DEFAULT CURRENT_TIMESTAMP;

ALTER TABLE products
MODIFY created_at DATETIME DEFAULT CURRENT_TIMESTAMP;

## <-----Q4----->

## 1. Add new row to countries
INSERT INTO countries
(code, name, continent_name)
VALUES
(1, 'Saudi Arabia', 'Asia');

## 2. Add new row to users
INSERT INTO users
(id, full_name, email, gender, date_of_birth)
VALUES
(1, 'Khalid', 'khalid@gmail.com', 'm', '1998-05-10');

## 3. Add new row to orders
INSERT INTO orders
(id, user_id, status)
VALUES
(1, 1, 'start');

## 4. Add new row to products

INSERT INTO products
(id, name, price, status)
VALUES
(1, 'Laptop', 3000, 'valid');

## 5. Add new row to order_products
INSERT INTO order_products
(order_id, product_id, quantity)
VALUES
(1, 1, 2);

## 6. Update row from countries
UPDATE countries
SET name = 'KSA'
WHERE code = 1;

## 7. Delete row from products
/*DELETE FROM products
WHERE id = 1;*/
-- step one delete record from order_product
DELETE FROM order_products
WHERE product_id = 1;

DELETE FROM products
WHERE id = 1;

## insert data
```

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```
INSERT INTO products (id, name, price, status)
VALUES (2, 'Mouse', 100, 'valid');

INSERT INTO products (id, name, price, status)
VALUES (3, 'laptop', 1200, 'expired');

## select
SELECT *
FROM products
WHERE id IN (
    SELECT product_id
    FROM order_products
);

SELECT * FROM users
```